

# Juliana Restrepo Tobar

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[github.com/jrestrepot](https://github.com/jrestrepot)

## TECHNICAL SKILLS

**Programming Languages**  
**Frameworks/Tools**  
**Languages**

**Expert:** Python  
Pytorch, scikit-learn, RAPIDS, keras, AWS, dask, Linux, MLflow  
**Experience in:** Java, Julia, SQL, MATLAB, R.  
Spanish (Native), English (Advanced-C2), French (Advanced-B2)

## EDUCATION

**EAFIT University, Medellín**

Bachelor of Mathematical Engineering, Minor in Computer Science

GPA: (4.56 / 0.0-5.0)

## EXPERIENCE

**Data Scientist at FinCrime Dynamics**

*Jan 2023 - Present*

- Developed "Scam Hunter", a product that enables users to simulate authentic Authorized Push Payment Fraud transactions, providing a realistic testing environment for fraud detection mechanisms.
- Developed "Case Reader", a product that reads fraud case studies and extracts key information using LLMs to build a knowledge graph, allowing users to easily manage cases and query information related to different entities.
- Developed a framework with specialized metrics and plots, enabling systematic comparison and validation of real-world data against synthetic data, thus heightening client confidence in our product's accuracy and reliability.
- Implemented diverse time series engines using generative Machine Learning algorithms, resulting in a significant improvement in our synthetic time series quality (32.4% from baseline).
- Implemented automated hyperparameter tuning processes for our AI models and defined metrics for synthetic tabular data, streamlining model optimization and enhancing the overall performance of the system.

**Researcher in Research Sprint at EDGELANDS Institute**

*Jan 2022 – Jun 2022*

- Collaborated with a multidisciplinary team to design predictive models to assess the probability of homicides in Medellín, using information from emergency calls, as well as geographic and socioeconomic variables.
- Implemented various supervised models that predict the probability of a homicide given an emergency call, enabling law enforcement agencies to allocate resources efficiently and prioritize response efforts to areas with higher predicted probabilities of homicides.

**Research Assistant at EAFIT University**

*Jul 2022 – Dec 2022*

- Implemented a highly accurate predictive model for electric vehicle energy consumption, aiming to identify optimal charging stations and create an app to alleviate "range anxiety" in EV owners.

**Student Representative at EAFIT**

*Aug 2021 – Jun 2022*

- Engaged in the university's internal dialogues, developed strategies to tackle the mental health epidemic amongst the students, and mediated assemblies for student discussions.

**Academic Tutor, Logic and Data Structures and Algorithms at EAFIT**

*Jan 2021 – Dec 2021*

- Led weekly hours of attention to 10+ students, helping them to understand the topics of the subjects.

## COURSES AND CERTIFICATIONS

- [Getting Started with Deep Learning](#)
- [Fundamentals of Accelerated Computing with CUDA Python](#)
- [Accelerating End-to-End Data Science Workflows](#)

## PROJECTS

- Unsupervised Learning Repository [\[GitHub\]](#)
- Relationship between crime and characteristics of urban parks in Medellín, Colombia [\[GitHub\]](#)
- Fire Detection Using Computer Vision [\[GitHub\]](#)
- Mesh Optimization Using NSGA-II [\[GitHub\]](#)

## RELEVANT UNIVERSITY COURSES

|                         |                                                  |                                                       |                         |
|-------------------------|--------------------------------------------------|-------------------------------------------------------|-------------------------|
| Optimization            | Heuristics                                       | Data Structures and Algorithms                        | Discrete Maths          |
| Multivariate Statistics | Probability Theory                               | Functional Analysis                                   | Artificial Intelligence |
| First Order Logic       | Robust and non-parametric statistical techniques | Advanced Multivariate Statistics for Machine Learning | Data Bases              |